



MES ABASAHEB GARWARE COLLEGE,PUNE
(AUTONOMOUS)

(Affiliated to Savitribai Phule Pune University)

First Year MSC. (CS) SEMESTER – 2

SUBMITTED BY:

SHARVARI SHUKLA (16646)

MADHURA DESHMUKH (16629)

PROJECT ON

FACE RECOGNITION ATTENDANCE SYSTEM

PROJECT GUIDED BY: DR. MANISHA BHARAMBE

INDEX

SR.NO	CONTENTS	PAGE NO.
1	Acknowledgement	1
2	Introduction	2
3	Existing System	3-5
4	Scope of the System	6
5	Features of the System	7
6	System Requirement • Hardware • Software	8
7	Feasibility study • Technical Feasibility • Economical Feasibility • Operational Feasibility	9
8	Analysis and Design • ER Diagram • UML Diagram	10-15
9	Data Dictionary	16-17
10	Input-Output Screen	18-20
11	Limitations/Drawbacks	21
12	Future Enhancement	22
13	Bibliography	23

1. Acknowledgement

We would like to express our sincere gratitude to our project guide, **Mrs. Manisha Bharambe**, for her valuable guidance, continuous support, and encouragement throughout the development of our project titled “*Face Recognition Attendance System.*” Her expertise and insights helped us at every stage of the project.

We are also thankful to our institution for providing us with the necessary resources and environment to successfully complete this project.

We would like to express our appreciation for the teamwork, cooperation, and dedication shared between us, which made this project successful and an enriching learning experience.

Lastly, we would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

Sharvari Shukla (16646)

Madhura Deshmukh (16629)

2. Introduction

In today's digital era, automation has become an essential part of various sectors, including education and corporate environments. One of the important aspects that require automation is attendance management. Traditional attendance systems, such as manual registers and IDbased methods, are time-consuming, inefficient, and prone to errors and fraudulent practices like proxy attendance.

To overcome these limitations, the **Face Recognition Attendance System** has been developed as a smart and automated solution. This system uses advanced technologies such as **Computer Vision** and **Machine Learning** to identify individuals based on their facial features and automatically mark their attendance.

Face recognition is a biometric technology that captures and analyzes unique facial characteristics of an individual. In this project, a webcam is used to capture real-time images, and these images are processed using algorithms to detect and recognize faces. Once a face is successfully recognized, the system records the attendance along with the date and time, ensuring accuracy and reliability.

The system is implemented using **Python programming language** along with powerful libraries such as **OpenCV**, **NumPy**, and **face recognition library**. These tools enable efficient image processing, face detection, encoding, and comparison with stored datasets.

The major objective of this project is to create a system that is:

- **Automated** – Eliminates manual intervention
- **Accurate** – Reduces human errors
- **Secure** – Prevents proxy attendance
- **Efficient** – Saves time and effort

This system can be widely used in various domains such as educational institutions, offices, and organizations where maintaining accurate attendance records is crucial.

Additionally, the project provides a user-friendly interface and ensures that attendance data is stored systematically, in CSV files and databases, making it easy to access, manage, and analyze.

In conclusion, the Face Recognition Attendance System is a modern, efficient, and reliable solution that leverages emerging technologies to improve traditional attendance methods and contributes towards digital transformation.

3. Existing System

Before the implementation of modern automated systems, attendance management was primarily handled using traditional methods. These systems are still widely used in many educational institutions, offices, and organizations. However, they come with several limitations that affect efficiency, accuracy, and security.

3.1 Manual Attendance System

The most common traditional method is the **manual attendance system**, where attendance is recorded using paper registers. In this method, a teacher or authorized person calls out names or passes around a register for individuals to sign.

Working:

- Attendance is marked by writing names or signatures in a register
- The record is maintained physically
- Attendance is calculated manually at the end of a period or month

Limitations:

- Time-consuming, especially for large groups
- High chances of human error
- Easy to manipulate (proxy attendance)
- Difficult to maintain and manage records
- No real-time data tracking

3.2 ID Card-Based Attendance System

In this system, students or employees use ID cards (RFID or barcode-based) to mark their attendance.

Working:

- Each individual is assigned a unique ID card
- The card is scanned using a device
- Attendance is recorded in the system

Limitations:

- Proxy attendance possible (one person can carry another's ID)

- Cards can be lost, damaged, or forgotten
 - Requires additional hardware (RFID scanners)
 - Maintenance cost is higher
-

3.3 Biometric Attendance System (Fingerprint)

Biometric systems use physical characteristics such as fingerprints to identify individuals.

Working:

- User places finger on biometric device
- System matches fingerprint with stored data
- Attendance is recorded automatically

Limitations:

- Requires physical contact (not hygienic)
 - Devices can fail due to dirty or damaged fingerprints
 - High installation and maintenance cost
 - Slower when many users are present
 - Limited scalability
-

3.4 Semi-Automated Computer-Based Systems

Some systems use software where attendance is entered manually into a computer system.

Working:

- Teacher inputs attendance into software
- Data is stored digitally

Limitations:

- Still requires manual effort
 - Chances of incorrect data entry
 - No automation in identification
-

3.5 Problems in Existing Systems

The major issues observed in traditional attendance systems include:

- Lack of automation

- Time-consuming process
- Proxy or fake attendance
- Data management difficulties

Conclusion of Existing System

From the above analysis, it is clear that traditional attendance systems are not efficient enough to meet the needs of modern institutions. They lack automation, security, and realtime capabilities.

These limitations highlight the need for an advanced system that is fast, reliable, and secure. This leads to the development of the **Face Recognition Attendance System**, which overcomes these drawbacks by using artificial intelligence and computer vision technologies.

4. Scope of the System

The **Face Recognition Attendance System** has a broad scope in environments where accurate, efficient, and automated attendance management is required. With the advancement of technologies like **Artificial Intelligence** and **Computer Vision**, this system can replace traditional attendance methods and improve overall productivity.

Application Areas

The system can be implemented in various sectors such as:

- Educational institutions like schools, colleges, and universities
- Corporate offices for employee attendance tracking
- Industries and organizations for workforce management
- Coaching classes and training centers

Functional Scope

The system performs several important functions, including:

- Capturing images through a webcam
- Detecting and recognizing faces using stored data
- Automatically marking attendance with date and time
- Storing records in a structured format (CSV or database)
- Providing real-time attendance processing

Technical Scope

The system is developed using **Python**, along with libraries such as **OpenCV** and face recognition modules. It can be further integrated with databases or cloud platforms and supports enhancements using advanced AI/ML techniques for better accuracy.

Future Scope

The system can be expanded with additional features such as:

- Mobile application support
- Cloud-based attendance management
- Mask detection capability
- Real-time notifications and advanced reporting

Conclusion

Overall, the Face Recognition Attendance System offers a modern solution for attendance management by improving accuracy, reducing manual effort, and enhancing security. It also provides flexibility for future upgrades and wider applications.

5. Features of the System

The **Face Recognition Attendance System** includes several advanced and useful features that make it more efficient, accurate, and secure compared to traditional attendance methods.

5.1 Face Detection

The system uses a webcam to capture images and detect human faces in real-time using computer vision techniques. It identifies the presence of a face within the camera frame.

5.2 Face Recognition

Once a face is detected, the system compares it with the stored dataset of registered users. If a match is found, the person is identified successfully.

5.3 Automatic Attendance Marking

After successful recognition, the system automatically marks attendance without any manual input. This reduces time and effort.

5.4 Date and Time Recording

The system records attendance along with the current date and time, ensuring proper tracking and record maintenance.

5.5 Data Storage

Attendance data is stored in a structured format such as CSV files or a database. This makes it easy to access, manage, and analyze records.

5.6 Real-Time Processing

The system works in real-time, meaning it instantly detects and recognizes faces as they appear in front of the camera.

5.7 User-Friendly Interface

The system provides a simple and easy-to-use interface, making it accessible even for nontechnical users.

5.8 Contactless System

Since the system uses facial recognition, it eliminates the need for physical contact, making it more hygienic and safe.

5.9 High Accuracy

With proper training of the dataset, the system provides high accuracy in identifying individuals.

Conclusion

These features make the Face Recognition Attendance System a smart, reliable, and efficient solution for modern attendance management.

6. System Requirements

The successful implementation of the **Face Recognition Attendance System** requires both hardware and software components. These requirements ensure smooth functioning, accuracy, and efficiency of the system.

6.1 Hardware Requirements

The basic hardware required for the system includes:

- **Computer/Laptop**
A system with minimum **4GB RAM** and a **dual-core processor (i3 or above)** is recommended for smooth execution.
- **Webcam**
A good quality webcam is necessary to capture real-time images for face detection and recognition.
- **Storage Device**
Sufficient storage is required to store datasets, images, and attendance records.

6.2 Software Requirements

The system is developed using the following software tools and technologies:

- **Python Programming Language**
Used for developing the entire system logic and functionality.
- **OpenCV Library**
Used for image processing, face detection, and video capturing.
- **NumPy**
Helps in handling arrays and mathematical operations efficiently.
- **Face Recognition Library (Tkinter, Pillow)**
Used for encoding and comparing facial features and Haarcascade algorithm.
- **IDE (Integrated Development Environment)**
Such as **VS Code, PyCharm, or Jupyter Notebook** for writing and executing code.
- **Operating System** Windows, Linux, or macOS

Conclusion

The system does not require high-end hardware or expensive software, making it cost-effective and easy to implement. With basic system configuration and open-source tools, the project can be successfully developed and deployed.

7. Feasibility Study

Feasibility study is an important step in system development that determines whether the proposed system is practical and beneficial. The **Face Recognition Attendance System** is analyzed based on different types of feasibility as follows:

7.1 Technical Feasibility

The system is technically feasible as it uses widely available technologies such as **Python**, **OpenCV**, and face recognition libraries. These tools are easy to use, well-documented, and suitable for developing computer vision applications.

The required hardware, such as a computer and webcam, is also easily available. Hence, the system can be developed and implemented without any major technical difficulties.

7.2 Economical Feasibility

The system is cost-effective as it does not require expensive hardware or licensed software. It uses open-source tools and libraries, which reduces development cost.

Only basic equipment like a webcam and a standard computer is needed, making it affordable for schools, colleges, and small organizations.

7.3 Operational Feasibility

The system is user-friendly and easy to operate. It reduces manual work and saves time, making it highly efficient for daily use.

Users do not require advanced technical knowledge to use the system, and attendance can be marked automatically with minimal effort.

7.4 Social Feasibility

The system promotes a contactless and hygienic method of attendance, which is especially beneficial in situations like pandemics.

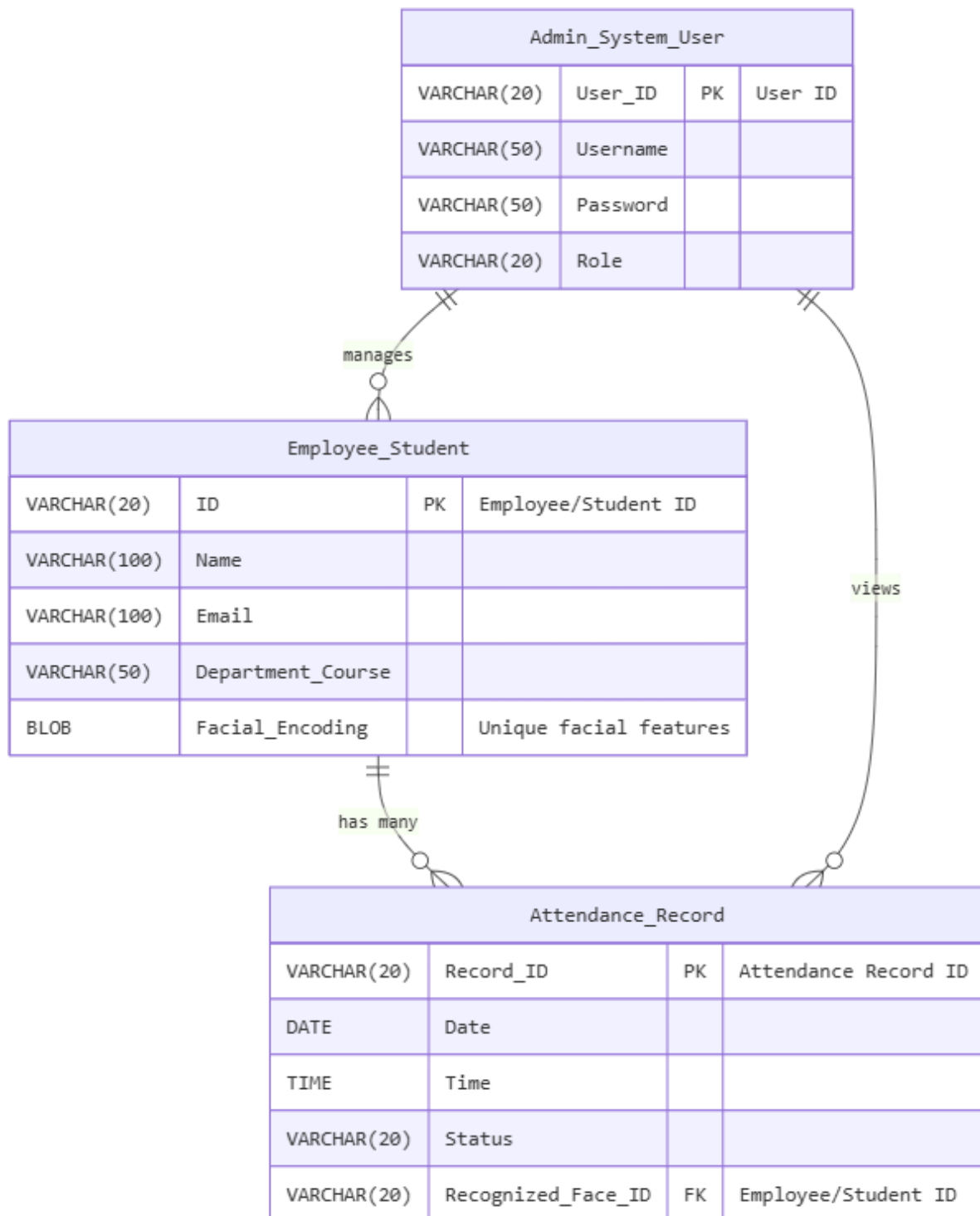
It also improves transparency and fairness by preventing proxy attendance.

Conclusion

From the above analysis, it is clear that the Face Recognition Attendance System is technically, economically, and operationally feasible. It is a practical and efficient solution for modern attendance management.

8. Analysis and Design

The **Analysis and Design** phase focuses on understanding how the system works and how its components are structured. It includes system flow, data handling, and interaction between different parts of the system.



1.

Flowchart

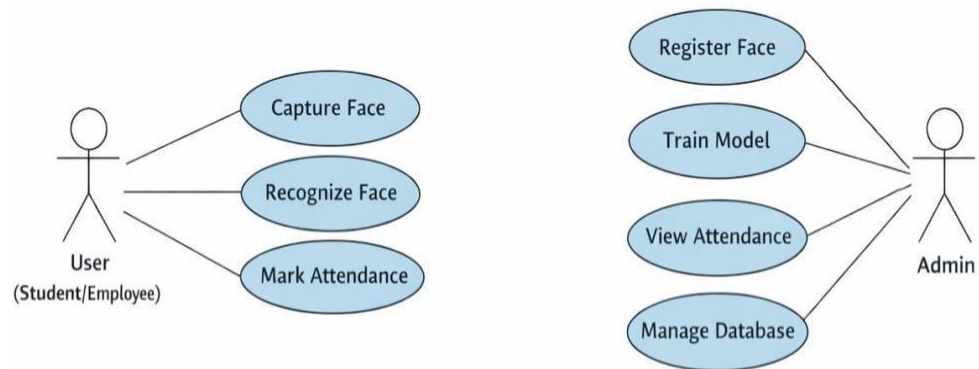
Face Recognition Attendance System Workflow



2.

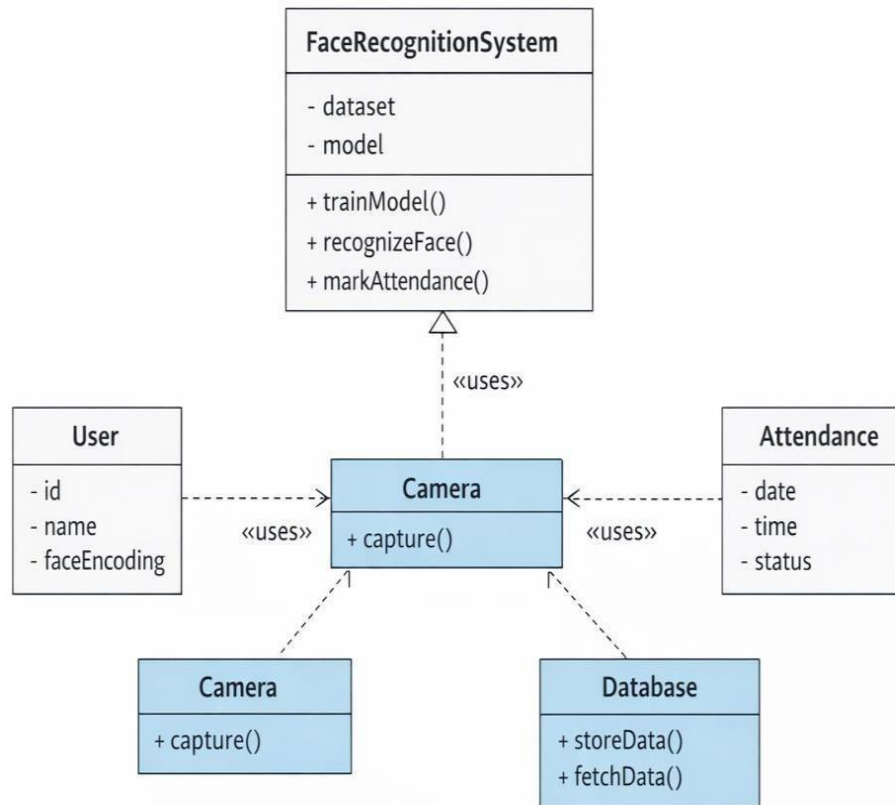
Use Case

Face Recognition Attendance System - Use Case Diagram



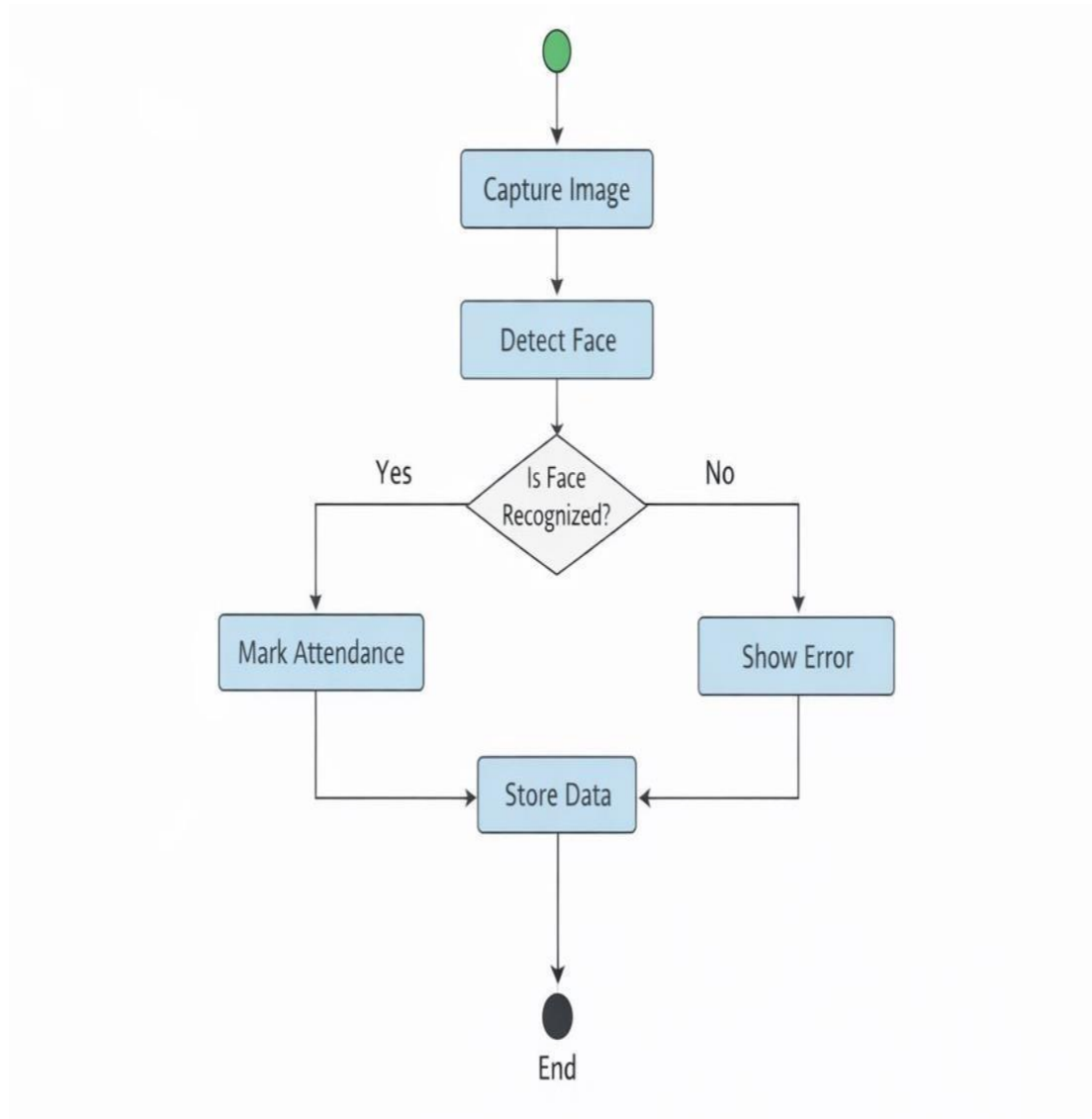
3.

Class Diagram

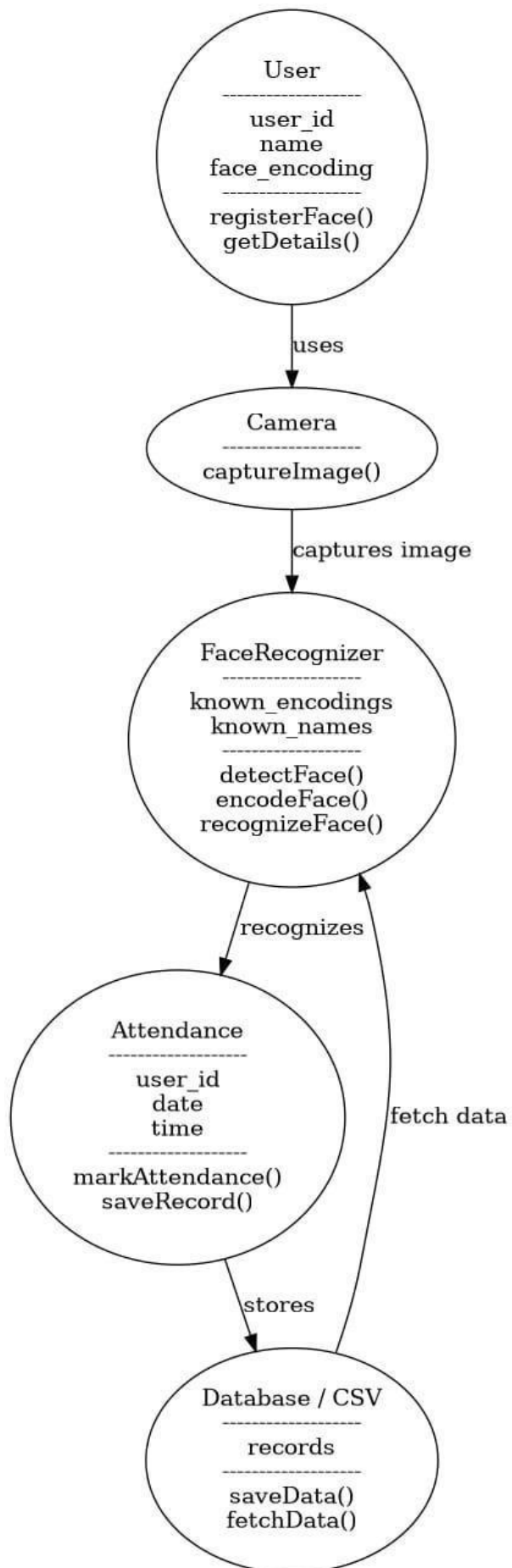


4.

Activity Diagram



UML diagram



9. Data Dictionary

A Data Dictionary is a structured description of all the data elements used in the system. It defines each field, its type, size, and purpose. In the Face Recognition Attendance System, the data dictionary helps in understanding how data is stored, managed, and processed.

9.1 Student Table

Field Name	Data Type	Size	Description
Student_ID	Integer	-	Unique identifier for each student (Primary Key)
Name	Varchar	50	Full name of the student
Roll_No	Varchar	20	Unique roll number assigned to student
College_Name	Varchar	100	Name of the college/institution
Email	Varchar	100	Email address of the student
Phone_No	Varchar	15	Contact number of the student
DOB	Date	-	Date of birth of the student
Course	Varchar	50	Course enrolled by the student
Semester	Integer	-	Current semester of the student
Year	Integer	-	Academic year (1,2,3,4)

9.2 Attendance Table

Field Name	Data Type	Size	Description
Attendance_ID	Integer	-	Unique identifier for each attendance record (Primary Key)
Student_ID	Integer	-	References Student_ID from Student table (Foreign Key)
Name	Varchar	50	Name of the student
Roll_No	Varchar	20	Roll number of the student
Date	Date	-	Date on which attendance is recorded
Time	Time	-	Time at which attendance is marked
Status	Varchar	10	Attendance status (Present/Absent)

9.3 Description of Key Fields

- **Primary Key (PK):**

A field that uniquely identifies each record in a table. Example:
Student_ID, Attendance_ID

- **Foreign Key (FK):**

A field used to link two tables together.

Example: Student_ID in Attendance table refers to Student table

9.4 Data Relationships

- One Student can have multiple Attendance records
- Relationship Type: **One-to-Many (1:M)**
- This ensures proper linking and avoids data redundancy

9.5 Data Integrity Constraints

To maintain accuracy and consistency, the following constraints are applied:

- **Uniqueness:** Student_ID and Roll_No must be unique
- **Not Null:** Important fields like Name, Student_ID, Date must not be empty
- **Valid Data Types:** Each field must store appropriate type (e.g., Date, Time)
- **Referential Integrity:** Student_ID in Attendance must exist in Student table

9.6 Purpose of Data Dictionary

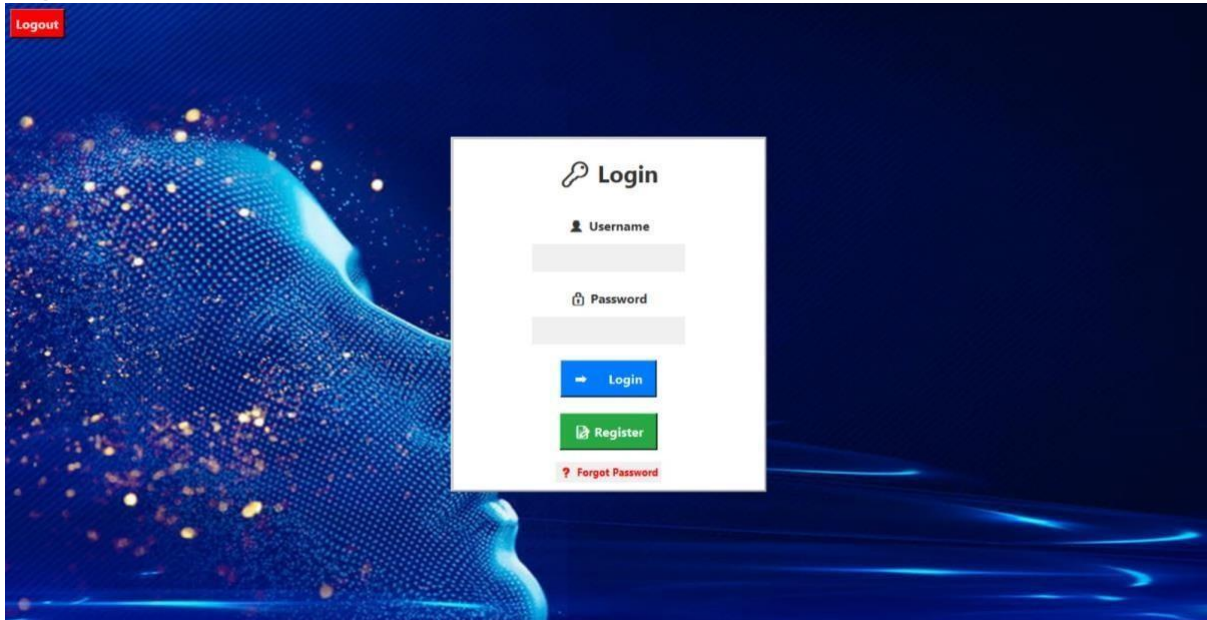
- Helps in understanding database structure
- Ensures consistency in data usage
- Improves system design and maintenance
- Assists developers and users in data handling

Conclusion

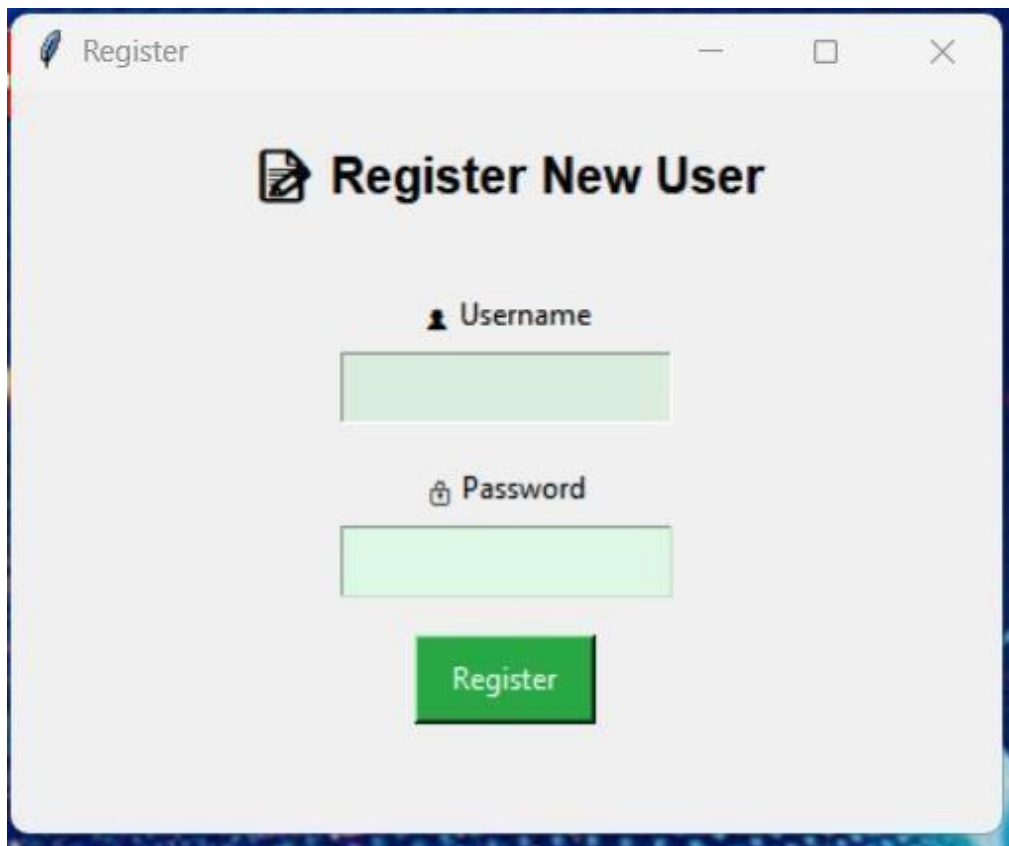
The Data Dictionary provides a clear and organized description of all data elements used in the Face Recognition Attendance System. It ensures proper data management, improves system efficiency, and supports accurate attendance tracking.

10. Input-Output Screen

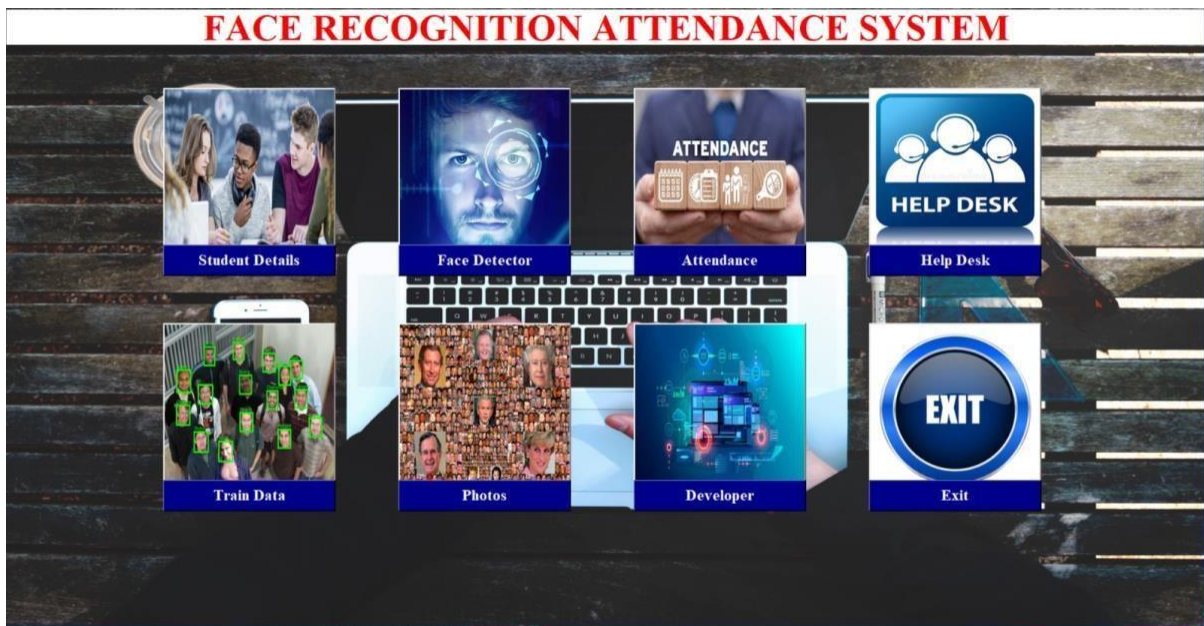
Login screen:



Register screen:



Dashboard:



Student details :

Student Management System EXIT

Student Form

Student ID

Roll

Name

Section

College

Department

Course

Semester

Year

Phone

Email

Address

Gender

DOB

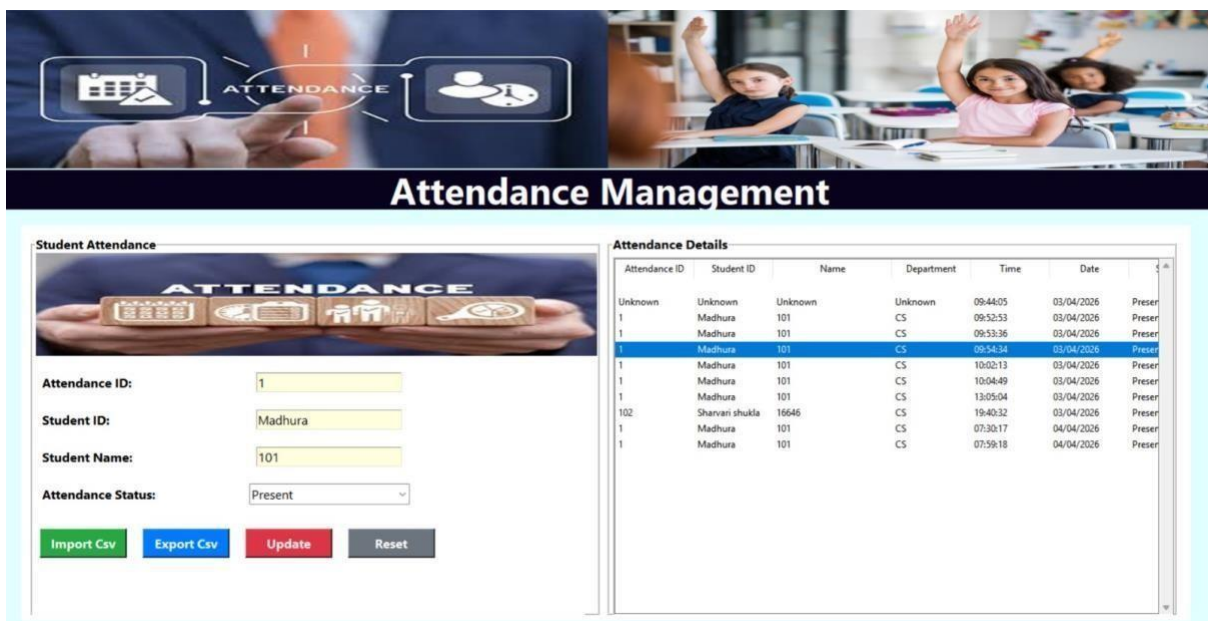
Student Records

Student ID	Roll	Name	Section	College	Department	Course	Semester	Year
1	101	Madhura	A	Garware	CS	M.Sc	2	1st
102	16646	Sharvari shukla	CS	Garware	CS	M.Sc	2	1st

Face detection:



Attendance Management:



11. Limitations / Drawbacks

The Face Recognition Attendance System is an efficient and reliable solution; however, it has a few minor limitations that can be improved in future versions.

- The system performs best under proper lighting conditions. In low-light environments, the accuracy of face detection may slightly decrease.
- The effectiveness of the system depends on the quality and variety of the stored dataset. Limited or unclear images may affect recognition performance.
- Variations in appearance such as wearing glasses, masks, or changes in hairstyle may impact accuracy if such variations are not included in the training data.
- The current system focuses on core functionality and may not include advanced security features like liveness detection or anti-spoofing.
- When handling a very large number of users, the system may experience a slight increase in processing time, which can be optimized further.

Conclusion

These limitations are minimal and do not significantly affect the overall performance. With future improvements, the system can become more accurate, secure, and scalable.

12. Future Enhancement

The **Face Recognition Attendance System** can be further improved with advanced features to enhance its functionality and performance. • Integration with mobile applications for easy access

- Cloud-based storage for real-time data management
- Advanced security features like liveness detection and anti-spoofing
- Mask detection for better real-world usability
- Real-time notifications via SMS or email
- Analytics dashboard for reports and insights
- Support for multiple users and locations

Conclusion

These enhancements can make the system more efficient, secure, and scalable for realworld applications.

13. Bibliography

The following resources were referred to during the development of the **Face Recognition Attendance System**:

- Python Official Documentation
- OpenCV Official Documentation
- Face Recognition Library Documentation
- NumPy Documentation
- Research papers on Face Recognition and Computer Vision
- Online tutorials and educational websites

Websites

- <https://docs.python.org>
- <https://opencv.org>
- <https://pypi.org/project/face-recognition/>
- <https://numpy.org>

Conclusion

These references provided the necessary knowledge and guidance for understanding and implementing the system effectively.

